

PROJECT REPORT

Credit Card Financial Analytics

Revenue drivers, spending behaviour and delinquency exposure: business insights and recommendations

Item	Detail
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Report date	July 2026
Analysis period	1 January 2023 to 31 December 2023 (53 reporting weeks)
Scope	10,293 credit card accounts; 15 customer attributes and 18 card and transaction attributes
Tools	Python (Pandas, NumPy, Matplotlib), MySQL (star schema, window functions, CTEs), Power BI (2-page dashboard)

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Appendix A. SQL business questions and results

1. Executive summary

This report analyses a 10,293-account credit card portfolio to answer one question: **where does the portfolio make money, where does it leak money, and what should the business do next?** The work combines Python data preparation, a MySQL star-schema with ten business queries, and a two-page Power BI dashboard. Every figure in this report was recomputed from the four source files.

10,293 ACCOUNTS	\$56.5M TOTAL REVENUE (DASHBOARD)	\$11.0M FEE + INTEREST INCOME	6.06% DELINQUENCY RATE	57.5% 30-DAY ACTIVATION
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Headline findings

- Premium cards earn far more per acquisition dollar.** Platinum and Gold return \$27.75 and \$24.46 of fee and interest income per \$1 of acquisition cost, against \$10.33 for Blue. Acquisition cost is almost identical across tiers (about \$94 to \$98 per account), so the gap comes entirely from income per account.
- There is a large, untapped upgrade pool.** 1,012 Blue cardholders earn \$100,000 or more. They already generate \$2,159 of income per account, more than double the Blue average of \$996, yet hold only a \$10,983 average limit on a Blue card. Together, all Silver, Gold and Platinum accounts number just 909.
- Credit limits are mis-sized in the middle-income band.** Customers earning \$70k to \$100k hold the lowest average limit (\$5,178) and show the highest utilisation (36.5%) and the highest delinquency among income tiers (7.02%). 465 of the 468 accounts running at 80% utilisation or above are Blue cards.
- Delinquency is stable but cannot be predicted from demographics.** The portfolio rate is a healthy 6.06%. Delinquency is statistically unrelated to income, spend, age, utilisation or satisfaction (all correlations below 0.03 in absolute value). A few job and education segments run hot (Self-employed Doctorate 11.11%, Government Uneducated 9.13%), but these segments are small and should be monitored rather than acted on blindly.
- Onboarding is the clearest operational leak.** Only 57.5% of accounts activate within 30 days. The 4,379 accounts that did not activate absorbed \$422K of acquisition cost, which is 42.6% of the total acquisition budget.
- Two headline trends need careful reading.** The Q4 spend spike and the +28.8% week-over-week gain both come from uneven reporting periods (Q4 has 14 weeks; Week 52 has only 164 accounts). Adjusted for this, Q4 spend per account is the lowest of the year. See Pillar 1.

Priority actions

Priority	Action	Evidence
1	Run an upgrade campaign for Blue holders earning \$100k+	1,012 accounts; \$2,159 income per account
2	Re-size credit limits for Blue accounts in the \$70k to \$100k band and review 468 high-utilisation accounts	36.5% utilisation; \$5,178 average limit
3	Introduce Day 7, 14 and 21 activation nudges	\$422K of acquisition cost on non-activated accounts
4	Shift acquisition spend toward high-return segments (premium tiers, Businessman, White-collar)	Income per \$1 CAC: 16.7x vs 7.7x for Self-employed
5	Fix KPI definitions and normalise trend metrics in reporting	Two revenue definitions; uneven weeks

2. Project overview and objectives

Business problem

Credit card issuers earn from annual fees, interest on revolving balances and (indirectly) from spend volume, while carrying acquisition costs and credit losses. Leadership needs a clear view of which products, customer segments and behaviours create profit, and where risk and inefficiency are building up.

Objectives

- Quantify overall portfolio performance: spend, revolving balance, income and delinquency exposure.
- Compare card tiers on profitability relative to customer acquisition cost (CAC).
- Understand how customers spend (category, ticket size, payment channel) and how that changes by quarter.
- Identify customer segments with elevated credit risk or unusual credit utilisation.
- Measure onboarding effectiveness and portfolio concentration.
- Deliver an interactive Power BI dashboard so non-technical stakeholders can explore the same questions.

End-to-end workflow

Stage	Tool	What was done
1. Prepare	Python (Pandas)	Loaded master and incremental batches, aligned schemas, unioned, de-duplicated, validated, standardised text and dates, ran EDA with charts
2. Model and query	MySQL	Staging tables, then a star schema (dim_customer, fact_credit_card), integrity checks, ten business questions using joins, CTEs and window functions
3. Visualise	Power BI	Two-page interactive dashboard (Transaction Report and Customer Report) with slicers and DAX measures
4. Communicate	This report	Insights, recommendations and limitations written for a business audience

3. Data, methodology and metric definitions

3.1 Data sources

File	Rows	Content
customer.csv	10,108	Customer demographics (master batch, Weeks 1 to 52)
cust_add.csv	185	Customer demographics (incremental batch, Week 53)
credit_card.csv	10,108	Card, transaction and risk metrics (master batch)
cc_add.csv	185	Card, transaction and risk metrics (incremental batch, Week 53)
Unified	10,293	One customer row and one card row per client_num (no duplicates, no missing values)

Each customer appears **once**, in a single reporting week. The data is therefore a set of weekly cohorts of roughly 195 different customers, not a panel that follows the same customers over time. This matters when interpreting trends (see Section 8).

3.2 Data preparation (Python)

- **Schema alignment:** column names standardised to lowercase snake_case; total_trans_vol in the master file renamed to total_trans_ct to match the incremental file.
- **Union:** master and incremental batches appended ($10,108 + 185 = 10,293$ rows per table).
- **Quality checks:** zero duplicate client_num values and zero null values in either table.
- **Standardisation:** text fields trimmed and title-cased (customer table); trailing spaces removed from card fields such as payment channel ("Swipe ", "Chip "); week_start_date validated and formatted DD-MM-YYYY; numeric fields cast to float where needed.
- **EDA:** merged table (10,293 rows, 33 columns) used for delinquency distribution, delinquency by income tier and by personal loan status, spend share by expenditure type and portfolio share by card tier.

3.3 Data model (MySQL star schema)

Cleaned data was loaded into staging tables, then split into a customer dimension and a card fact table joined on client_num (one-to-one, enforced with a primary key and foreign key). Reconciliation queries confirmed matching row counts across staging, dimension and fact tables and zero orphan records in either direction.

Table	Type	Grain	Key columns
dim_customer	Dimension	One row per customer	client_num (PK), age, gender, dependents, education, marital status, state, job, income, contact channel, satisfaction score
fact_credit_card	Fact	One row per card account	client_num (PK, FK), card category, annual fees, activation_30_days, CAC, week and quarter, credit limit, revolving balance, spend, transaction count, utilisation, payment channel, expense type, interest earned, delinquency flag

3.4 Metric definitions

Two revenue measures appear in this project and they answer different questions. Both are used here and are always labelled.

Metric	Definition	Value
Fee and interest income	annual_fees + interest_earned (used in all SQL queries)	\$10.98M
Total revenue (dashboard)	annual_fees + interest_earned + total_trans_amt, the headline figure on the Power BI cards	\$56.52M
Transaction spend	Sum of total_trans_amt	\$45.53M
Revenue (income) per CAC dollar	Fee and interest income divided by customer_acq_cost	\$11.08
Delinquency rate	Delinquent accounts divided by total accounts	6.06%
30-day activation rate	Accounts with activation_30_days = 1 divided by total accounts	57.46%
Average ticket size	Total spend divided by total transaction count	\$68.24
High-utilisation account	Average utilisation ratio of 0.80 or above	468 accounts

Why two revenue measures?

Transaction spend is customer volume, not issuer income. The income-based measure (fees plus interest) is the better indicator of profitability and is used for tier, segment and CAC comparisons. The spend-inclusive measure is kept because it is the headline on the dashboard; segment shares are very similar under both.

4. Portfolio snapshot

Measure	Value	Measure	Value
Accounts	10,293	Annual fees	\$3.00M
Transaction spend	\$45.53M	Interest earned	\$7.98M
Transactions	667,234	Fee and interest income per account	\$1,067
Revolving balance	\$11.98M	Total acquisition cost	\$0.99M (avg \$96)
Delinquent accounts	624 (6.06%)	Average credit limit	\$8,642
30-day activation	57.5% (5,914 accounts)	Average utilisation	27.5%
Average customer age	46.3 years	Median income	\$44,776 (mean \$57,087)
Average satisfaction score	3.19 out of 5	Interest share of fee + interest income	72.7%

The portfolio is large in customer count but modest in income per account. It is dominated by Blue cards (91.2% of accounts), concentrated in five US states (94% of accounts) and funded mainly by interest income (about 73% of fee and interest income).

5. Business insights

Insights are organised into seven pillars. Each pillar lists the evidence, explains what it means for the business, and ends with a recommendation.

Pillar 1: Financial health and revenue momentum

What the data shows

- The portfolio generated **\$56.52M of total revenue** on the dashboard definition: \$45.53M of spend, \$7.98M of interest and \$3.00M of annual fees. On the income-only basis it generated **\$10.98M**, or \$1,067 per account.
- Interest is the engine of profitability, contributing **72.7%** of fee and interest income. Annual fees contribute the remaining 27.3%.
- Quarterly spend is flat at \$11.1M to \$11.7M. Q4 is highest in total (\$11.70M, 2,689 accounts) but contains **14 reporting weeks** against 13 in Q1 to Q3, because the incremental Week 53 batch falls in Q4.
- Once normalised, Q4 is the weakest quarter: spend per week is \$836K (Q3: \$881K) and spend per account is \$4,350 (Q3: \$4,517). The data does **not** show a Q4 seasonal uplift.
- The dashboard shows Week 53 revenue up 28.8% on Week 52. Week 52 had only 164 accounts against 185 in Week 53, and a typical week has 195. Per account, total revenue rose 14.1% (\$5,690 to \$6,495) while fee and interest income per account **fell** 8.4% (\$1,125 to \$1,030).

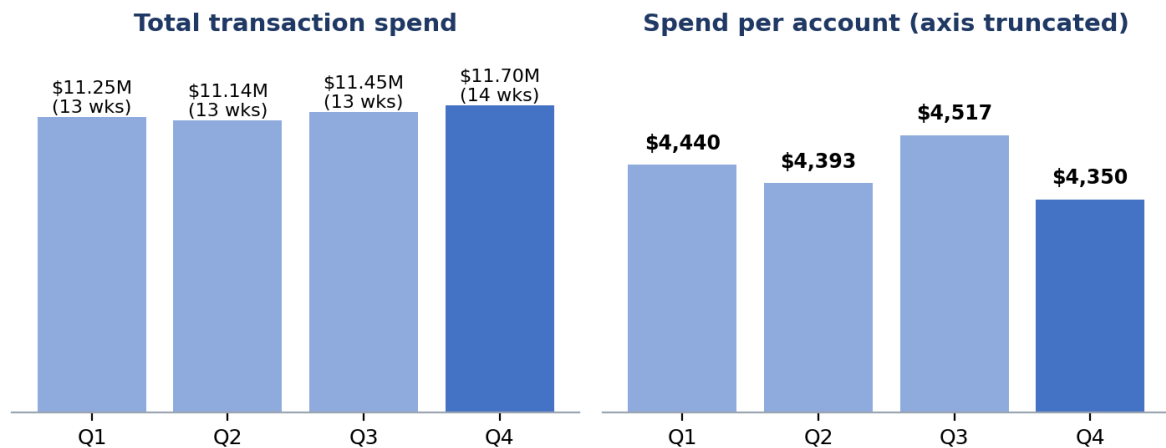


Figure 1. Quarterly transaction spend (left) and spend per account (right). Q4 contains 14 weeks.

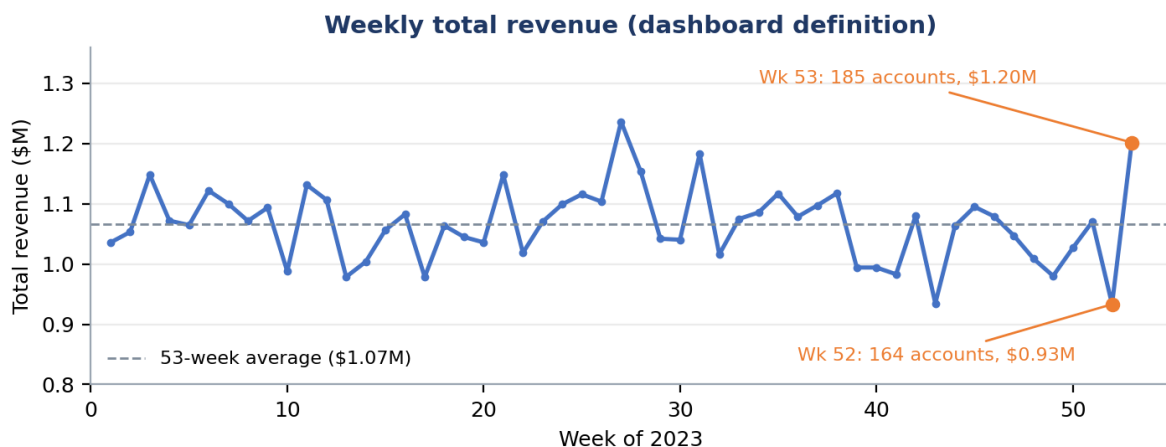


Figure 2. Weekly total revenue (dashboard definition). The Week 52 dip and Week 53 rebound reflect cohort size, not demand.

So what

Do not size seasonal campaigns on the raw Q4 total. Report spend per account and per week alongside totals, flag the Week 53 batch, and treat week-over-week swings with cohort size in mind. A reliable seasonality read needs several years of data.

Pillar 2: Card tier economics and acquisition efficiency

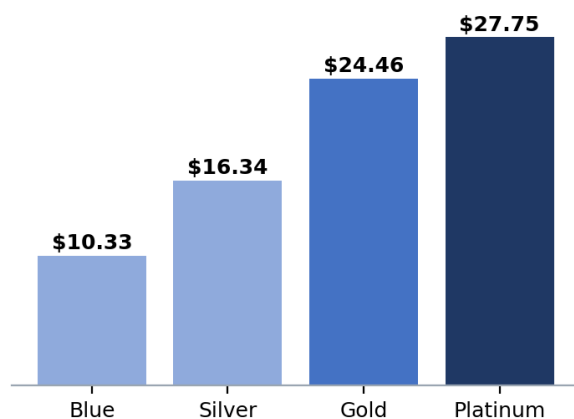
What the data shows

Tier	Accounts	Income	CAC	Net of CAC	Income per \$1 CAC	Income per account	Avg utilisation	Delinquency
Blue	9,384	\$9.35M	\$0.90M	\$8.44M	\$10.33	\$996	29.2%	6.05%
Silver	649	\$1.01M	\$0.06M	\$0.95M	\$16.34	\$1,559	8.3%	6.16%
Gold	193	\$0.44M	\$0.02M	\$0.42M	\$24.46	\$2,292	12.5%	6.22%
Platinum	67	\$0.18M	\$0.01M	\$0.18M	\$27.75	\$2,721	15.1%	5.97%

Income = annual fees + interest earned; net of CAC = income minus acquisition cost.

- Blue cards are the volume engine: **91.2% of accounts and 85.1% of income**. Blue and Silver together account for 93.3% of transaction spend.
- Acquisition cost per account is nearly identical across tiers (\$94 to \$98), so a Gold or Platinum account costs the same to win but earns **2.3x and 2.7x** as much as a Blue account (Silver: 1.6x).
- The 909 premium accounts (8.8% of the base) deliver 14.9% of income. Delinquency is flat across tiers (5.97% to 6.22%), so premium customers add income without adding observable risk.
- Premium tiers use little of their credit: utilisation is 8.3% (Silver), 12.5% (Gold) and 15.1% (Platinum) against 29.2% for Blue.
- Design inconsistencies stand out. Platinum has a lower average limit (\$16.5k) than Silver (\$23.5k) or Gold (\$21.8k), and annual fees barely rise with tier (Blue \$291, Silver \$292, Gold \$298, Platinum \$308).

Fee + interest income per \$1 of CAC



Account mix vs income mix

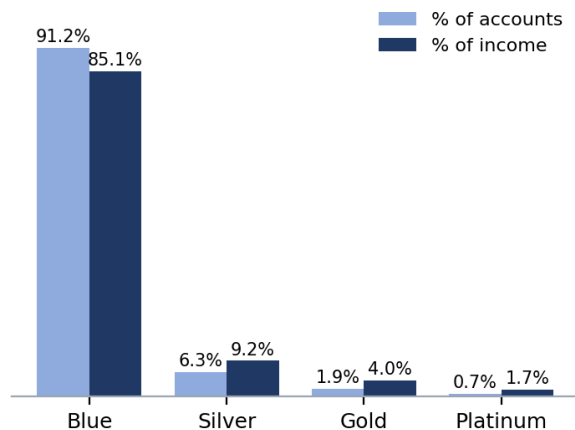


Figure 3. Income per \$1 of CAC by tier (left); share of accounts versus share of income (right).

So what and recommendation

Redirect part of top-of-funnel spend toward premium-eligible prospects and launch an **upgrade programme for the 1,012 Blue holders earning \$100k or more**. They already earn \$2,159 per account with 20.7% average utilisation, so they are engaged, solvent and under-served by a Blue product.

Review the tier proposition: align credit limits with tier hierarchy and test fee differentiation, since fees currently do not reward premium positioning. Note that Platinum is only 67 accounts, so its ratio should be validated on a larger base before scaling.

Pillar 3: Spending behaviour and payment channels

What the data shows

- Everyday categories anchor spend. **Bills is the largest category** (\$11.17M, 24.5%), followed by Entertainment (\$7.82M, 17.2%), Fuel (\$7.71M, 16.9%), Grocery (\$7.03M, 15.4%), Food (\$6.81M, 14.9%) and Travel (\$5.00M, 11.0%).
- **Travel has the highest average ticket at \$100.51** (portfolio average \$68.24) and the highest income per account among categories (\$1,447). Food is next on both measures (ticket \$75.86; \$1,295). Bills has the lowest income per account (\$941).
- **Swipe** carries 62.6% of spend across 7,232 accounts. **Chip** users spend the most per account (\$5,765 against \$3,944 for Swipe) and earn \$1,291 income per account, but show the highest delinquency at **7.24%**. **Online** is the smallest channel (6.3% of spend) with the lowest delinquency (4.64%) and the lowest activation rate (56.0%).

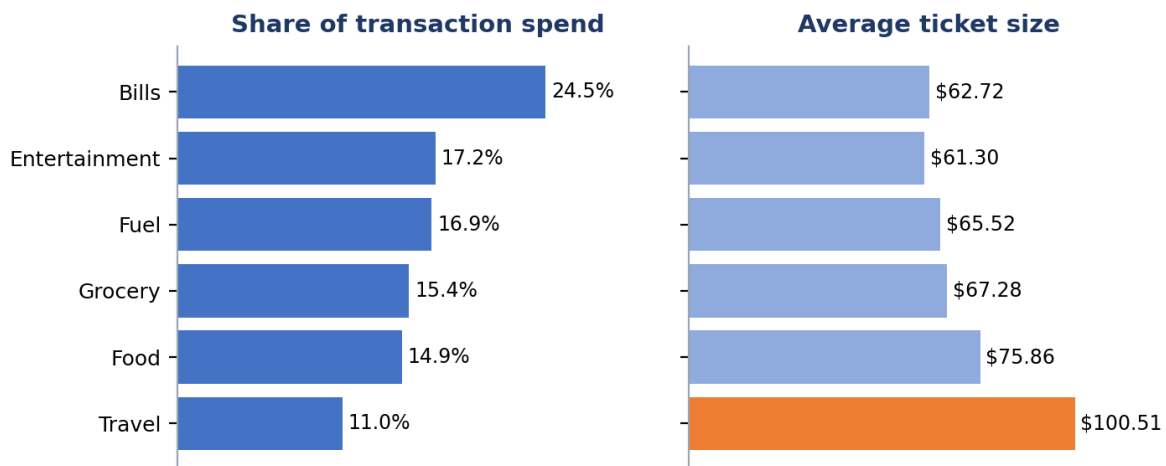


Figure 4. Share of spend (left) and average ticket size (right) by expenditure category.

Channel	Accounts	Spend share	Spend per account	Avg ticket	Delinquency	Income per account
Swipe	7,232	62.6%	\$3,944	\$63.86	5.78%	\$988
Chip	2,457	31.1%	\$5,765	\$78.37	7.24%	\$1,291
Online	604	6.3%	\$4,714	\$71.41	4.64%	\$1,108

So what and recommendation

Protect frequency in Bills and Fuel with targeted cash-back, and build **co-branded Travel and Food rewards** to win high-ticket, high-income spenders.

Treat Chip as a high-value but higher-risk channel: keep it, and pair it with tighter monitoring. Grow Online as a lower-risk channel, but fix its weak activation first.

Pillar 4: Credit risk, delinquency and utilisation

What the data shows

- **624 of 10,293 accounts (6.06%) are delinquent.** Their revolving balances total \$704K, 5.9% of the portfolio balance. Delinquency by quarter ranges from 5.36% (Q2) to 7.14% (Q1).
- Delinquency is **not explained by the variables available.** Its correlation with income, spend, utilisation, credit limit, age and satisfaction is between -0.02 and +0.01. Personal-loan holders (6.5% against 6.0%) and house owners (6.3% against 5.8%) differ only slightly.

- Job and education combinations do show spread. **Self-employed Doctorate holders run at 11.11%** (15 of 135) and **Government Uneducated at 9.13%** (20 of 219), against 6.06% overall. The lowest are Businessman High School (3.30%) and White-collar Doctorate (3.77%). With these sample sizes, the 95% range for the top segment is roughly 6.8% to 17.5%, so the result is a watch-list signal, not proof.

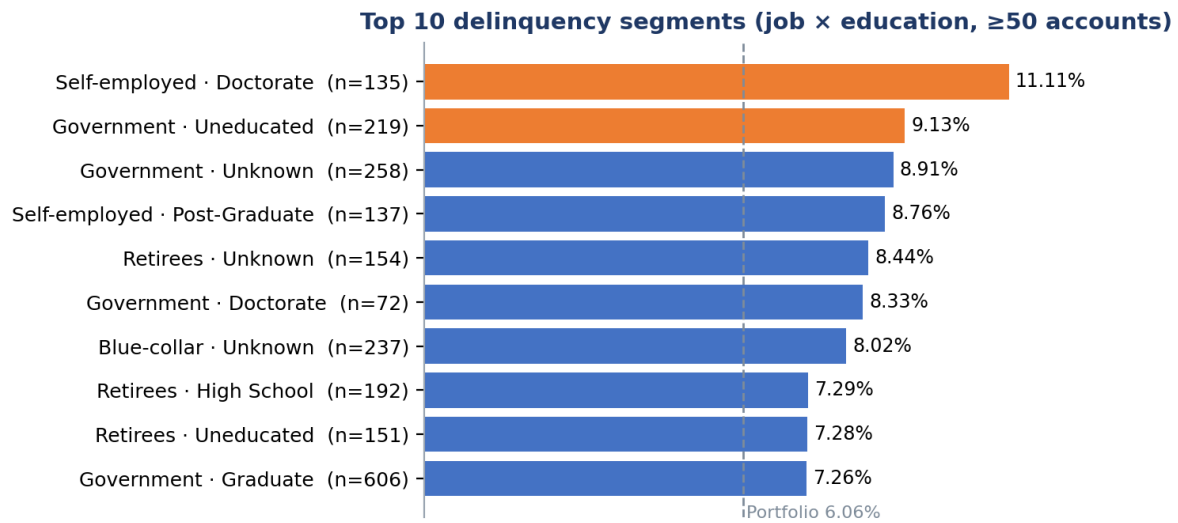


Figure 5. Highest-delinquency segments by occupation and education (minimum 50 accounts).

Utilisation and credit limits by income

Income tier	Accounts	Avg limit	Avg utilisation	Accounts at 80%+ utilisation	Delinquency
Under \$35k	4,064	\$8,433	25.9%	206	6.20%
\$35k to \$70k	3,201	\$7,859	29.9%	150	5.59%
\$70k to \$100k	1,510	\$5,178	36.5%	103	7.02%
\$100k to \$150k	956	\$14,657	17.2%	9	5.86%
\$150k and above	562	\$13,698	18.3%	0	5.52%

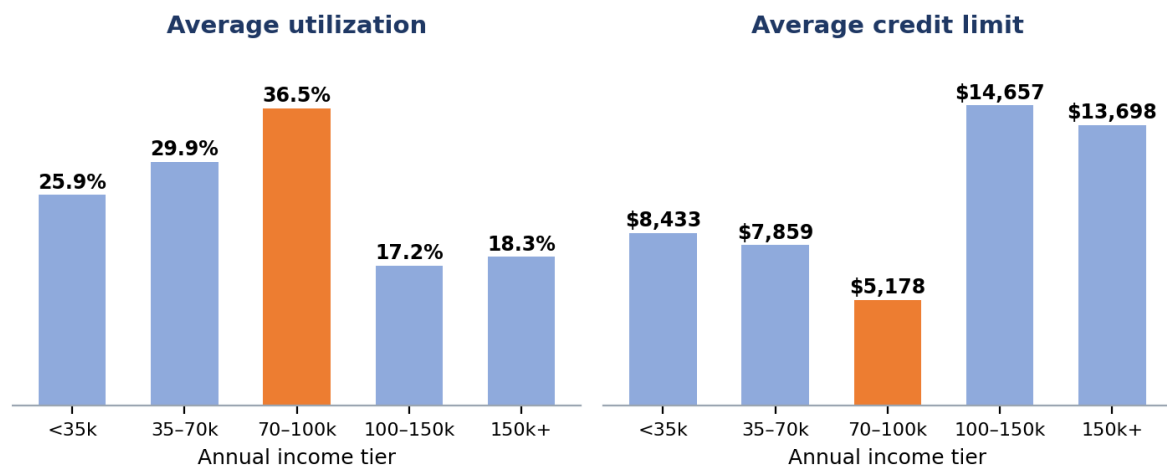


Figure 6. Utilisation and credit limit by income tier. The \$70k to \$100k band is the outlier.

- The \$70k to \$100k band is 98% Blue, with an average Blue limit of only \$4,704. That under-sizing pushes utilisation to 36.5% and delinquency to 7.02%, the highest of any income tier.
- Of the **468 accounts at 80% utilisation or above**, **465 are Blue**, with an average limit of just \$2,407. Interestingly, these accounts are delinquent at 4.3%, below the portfolio average, which suggests they are stretched but still paying.

- High earners (\$150k+) use only 18.3% of their limit and none are above 80%.

So what and recommendation

Launch a **credit-limit review for Blue accounts in the \$70k to \$100k band** and for the 468 high-utilisation accounts. Increase limits only where payment behaviour supports it, since higher utilisation here has not yet translated into higher delinquency.

Place the high-delinquency job and education segments on a **watch list** with enhanced early-warning monitoring rather than blanket tightening. Because demographics explain almost none of the delinquency, the next analytical step is to add payment history, days past due and bureau data to build a proper risk score.

Pillar 5: Customer segments, demographics and geography

What the data shows

Occupation	Accounts	Share of accounts	Share of income	Income per account	Income per \$1 CAC	Delinquency
Businessman	1,935	18.8%	28.8%	\$1,632	\$16.73	5.27%
White-collar	1,564	15.2%	17.5%	\$1,230	\$12.62	5.56%
Government	1,554	15.1%	14.9%	\$1,051	\$10.92	7.27%
Retirees	1,006	9.8%	8.4%	\$922	\$9.42	6.26%
Blue-collar	1,606	15.6%	13.1%	\$895	\$9.31	5.48%
Self-employed	2,628	25.5%	17.3%	\$724	\$7.69	6.51%

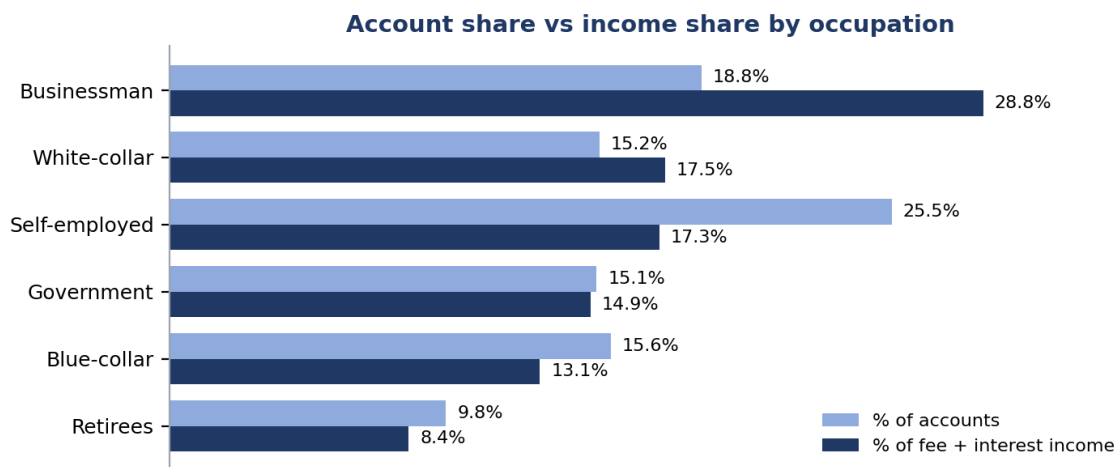


Figure 7. Share of accounts versus share of fee and interest income by occupation.

- **Businessman** customers are 18.8% of accounts but 28.8% of income. **Self-employed** customers are the largest group (25.5% of accounts) yet the least valuable per account (\$724) and the least efficient per acquisition dollar (\$7.69), with above-average delinquency (6.51%).
- Income is the dominant driver of value: income correlates 0.93 with spend, 0.82 with transaction count and 0.80 with interest earned. Credit limit, age and satisfaction add almost nothing.
- **Gender:** women are 58.2% of accounts but 49.3% of income (\$904 against \$1,294 per account). The gap follows earnings: median income is \$27.5k for women and \$76.6k for men, and 34% of women are Self-employed against 14% of men. Women activate more readily (59.0% against 55.3%).
- **Age:** customers aged 40 to 59 make up 74% of accounts, and the 50 to 59 group has the highest income per account (\$1,160). Under-30s are only 2.1% of accounts (220), and the 60 to 69 group earns the least among sizeable segments (\$855).
- **Dependents:** customers with 2 or 3 dependents are 53% of the base and contribute 54% of income.

- **Geography:** Texas (23.1% of income), California (23.1%), New York (22.9%), Florida (17.6%) and New Jersey (7.4%) generate **94.1%** of income. The remaining 21 states together contribute 5.9%.

So what and recommendation

Weight acquisition budgets toward Businessman and White-collar prospects and re-examine the economics of Self-employed acquisition (lower-limit starter products, lower CAC channels, or fee changes). Use the 50 to 59 profile as the lookalike model for premium targeting.

Treat the five-state concentration as both a strength (operational focus) and a risk (regional exposure). Any expansion test should be small and measured, since the tail states have too few accounts for reliable conclusions.

Pillar 6: Customer experience and satisfaction

What the data shows

- Average satisfaction is **3.19 out of 5**. 40.7% of customers score 4 or 5, while 29.3% score 1 or 2.

Satisfaction score	Customers	Income per account	Delinquency	Avg utilisation
1	1,203	\$1,098	6.3%	27.4%
2	1,812	\$1,068	6.1%	26.8%
3	3,090	\$1,061	6.2%	28.0%
4	2,154	\$1,075	5.8%	27.7%
5	2,034	\$1,048	6.0%	27.0%

- The SQL ranking places satisfaction score 1 first for income per account (\$1,098). However, the spread across all five scores is only about 5% (\$1,048 to \$1,098) and the correlation between satisfaction and income is effectively zero (-0.01).
- Delinquency and utilisation are also flat across scores. The data therefore **does not support** a story that unhappy customers are financially stressed or that satisfaction drives revenue.

So what and recommendation

Satisfaction is a service-quality metric here, not a revenue or risk predictor. Track it on its own terms: investigate why three in ten customers score 1 or 2, and add reasons-for-score and complaint data so it can be linked to attrition. Customer churn is not captured in this dataset, so any retention claim would need that field first.

Pillar 7: Onboarding friction and portfolio concentration

What the data shows

- **57.5% of accounts activate within 30 days;** 4,379 do not. Acquisition cost spent on those accounts is **\$422K, or 42.6% of total CAC.**
- By acquisition contact channel, activation is 61.0% for Unknown, 56.9% for Cellular and 53.9% for Telephone. By card tier it ranges from 61.9% (Silver) to 55.4% (Gold). Women activate at 59.0% against 55.3% for men.
- Non-activation in the first 30 days does not show up in revenue: accounts that did not activate earn \$1,079 per account against \$1,059 for those that did. Many of these customers evidently activate later, so the cost is delayed income and wasted acquisition spend, not lost accounts.

30-day activation rate

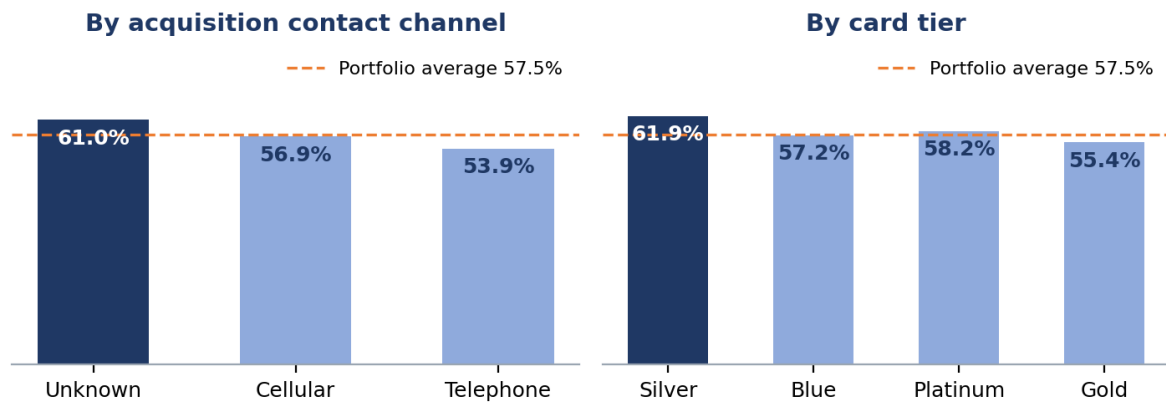


Figure 8. 30-day activation rate by acquisition channel and by card tier.

- **Spend is well diversified.** It takes **5,856 cardholders (56.9%)** to generate 80% of total spend, far above the classic 20% rule. The top 10% of cardholders contribute 30.0% of spend and the top 20% contribute 44.2%.

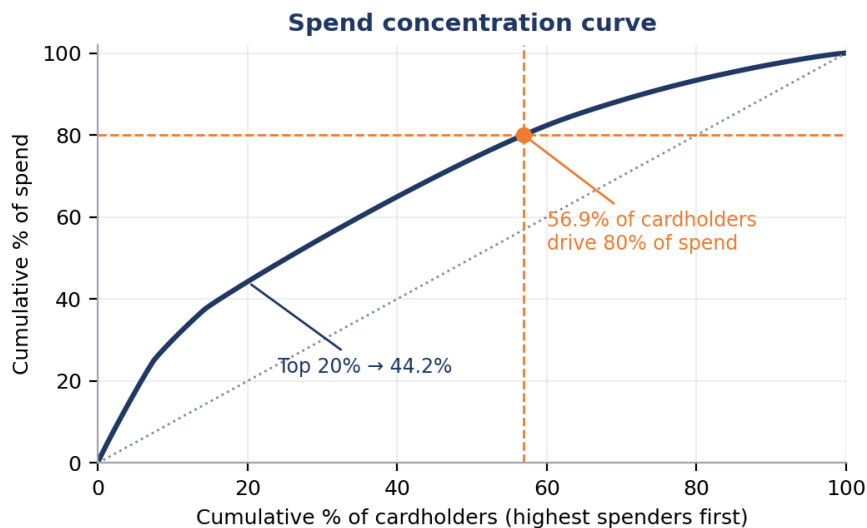


Figure 9. Spend concentration curve. The portfolio has low customer concentration risk.

So what and recommendation

Concentration risk is low, so the priority is efficiency, not defence. Deploy **automated SMS and email nudges on Day 7, Day 14 and Day 21** after approval and measure the lift against a holdout group. Each additional percentage point of activation is about 103 more activated accounts. Review why Telephone-sourced accounts lag, and whether the Unknown channel (the best performer) can be identified and replicated.

6. Power BI dashboard guide

The dashboard presents the cleaned customer and credit card tables in a two-page report with shared filters. It lets business users reproduce and extend the findings in this report without writing SQL.

Page 1: Credit Card Transaction Report

- **KPI cards:** Total Revenue, Transaction Amount, Total Interest, Transaction Count.
- **Visuals:** revenue by card category (table with interest and annual fees), quarterly revenue and transaction count (combo chart), revenue by expenditure type, education, customer job and payment channel (Use Chip), and customer acquisition cost by card category.
- **Slicers:** Quarter, Week Start Date, Gender, Salary Group, Card Category.

Page 2: Credit Card Customer Report

- **KPI cards:** Total Revenue, Total Income, Total Interest and customer satisfaction (CSS).
- **Visuals:** revenue trend by gender, gender treemap, top five states, age group, salary group, dependent count, education level, marital status and a job table combining revenue, spend and income.
- **Slicers:** Quarter, Week Start Date, Salary Group, Card Category.

Dashboard values to validate against

Measure	Value in the data
Total revenue (fees + interest + spend)	\$56.52M
Transaction amount / interest / annual fees	\$45.53M / \$7.98M / \$3.00M
Transaction count	667,234
Revenue by card: Blue / Silver / Gold / Platinum	\$47.19M / \$5.66M / \$2.53M / \$1.14M (83.5% / 10.0% / 4.5% / 2.0%)
Revenue by quarter: Q1 / Q2 / Q3 / Q4	\$13.96M / \$13.82M / \$14.24M / \$14.50M
Revenue by channel: Swipe / Chip / Online	63.1% / 30.7% / 6.2%
Revenue by gender: Male / Female	54.7% / 45.3%
Top five states by revenue	Texas, New York, California, Florida, New Jersey

Suggested dashboard improvements

- Add per-account and per-week measures beside every total, so uneven periods (Q4, Week 53) cannot mislead.
- Show both revenue definitions on the cards with clear labels, or rename the headline "Total Revenue incl. Spend".
- Add risk visuals: delinquency rate and 80%+ utilisation count by income tier and by job, and an activation-rate card by channel.
- Use state_cd for geography. Zip code is not reliable (see Section 8).

7. Consolidated recommendations and roadmap

#	Recommendation	Evidence	KPI to track	Priority
1	Upgrade campaign for Blue holders earning \$100k+	1,012 accounts; \$2,159 income per account; 20.7% utilisation	Upgrade conversion; income per account	High
2	Credit-limit review: Blue accounts at \$70k to \$100k income and 468 high-utilisation accounts	36.5% utilisation; \$5,178 avg limit; 465 of 468 are Blue	Utilisation; delinquency; limit uptake	High
3	Day 7, 14, 21 activation nudges with a holdout test	57.5% activation; \$422K CAC on non-activated accounts	30-day activation rate; CAC per activated account	High
4	Re-weight acquisition toward Businessman, White-collar and premium-eligible prospects	Income per \$1 CAC: 16.7x vs 7.7x (Self-employed)	Income per \$1 CAC by segment	High
5	Correct reporting: normalise by week and account; label revenue definitions	Q4 has 14 weeks; Week 52 has 164 accounts	Per-account and per-week trend KPIs	High
6	Watch-list and early-warning for high-delinquency segments; enrich risk data	Self-employed Doctorate 11.1%; Government Uneducated 9.1%	Segment delinquency; roll rates	Medium
7	Cash-back on Bills and Fuel; co-branded Travel and Food rewards	Bills 24.5% of spend; Travel ticket \$100.51	Spend per account; category share	Medium
8	Review tier design: limits and fees by tier	Platinum limit below Silver; fees \$291 to \$308 across tiers	Income by tier; premium uptake	Medium
9	Grow Online, fix its activation; monitor Chip risk	Online 4.64% delinquency, 56.0% activation; Chip 7.24% delinquency	Channel mix; channel delinquency	Medium
10	Small-scale test for under-30 and out-of-footprint acquisition	Under-30s 2.1% of accounts; 21 states = 5.9% of income	Test-cell income per account	Low

Suggested sequencing

- **Next 30 days:** fix KPI definitions in reporting; launch the activation nudge test; pull the Blue \$100k+ and high-utilisation lists.
- **30 to 90 days:** run the upgrade pilot and the limit review; set up watch-list monitoring.
- **90 days onward:** redesign tier proposition; add payment-history data and build a delinquency score; evaluate acquisition re-weighting.

8. Limitations and data notes

Good analysis states what it cannot show. The following points affect how far the findings can be pushed.

- **Cross-sectional data.** Each customer appears once in one week. Week-to-week and quarter-to-quarter movements compare different customers, so they indicate patterns rather than customer-level trends. One year of data cannot establish seasonality.
- **Uneven periods.** Q4 contains 14 weeks (Week 53 is an incremental batch of 185 accounts), and Week 52 holds only 164 accounts against a typical 195.
- **Revenue definition.** The dashboard headline includes transaction spend; the SQL analysis uses fees plus interest. Both are valid but not interchangeable.
- **Interest field behaviour.** Interest earned tracks income and spend (correlation 0.80 and 0.81) rather than revolving balance (0.06), and total interest equals 67% of total revolving balance. This is unusual for a real portfolio and should be validated against source systems before it drives pricing decisions.
- **Geography.** 97.8% of records carry the same zip code (91750), regardless of state, so zip code cannot be used. State code is used instead.
- **Small segments.** Platinum (67 accounts), most states outside the top five, and several job and education cells have small samples. Percentages there are volatile.
- **Unknown values.** Contact channel is Unknown for 19.2% of accounts, education for 15.0% and marital status for 7.4%.
- **Risk data depth.** Delinquency is a single binary flag with no payment history, days past due, balances over time or bureau data. Correlation does not imply causation.
- **No attrition field.** The dataset does not record closed accounts or churn, so retention conclusions are outside scope.

9. Conclusion

The portfolio is healthy and well diversified: delinquency is a stable 6.06%, spend is spread across the customer base, and income per acquisition dollar is strong. The biggest opportunities are not in finding new customers but in **getting more from the ones already acquired**: upgrading high-income Blue holders, right-sizing limits in the middle-income band and activating the 42.5% of accounts that stall in the first 30 days.

The biggest analytical lesson is about measurement. Uneven reporting periods and two revenue definitions can both produce misleading headlines, so normalised, clearly labelled KPIs are essential. With payment history added to the dataset, the next step is a predictive delinquency model and a controlled test of the upgrade and activation programmes.

Appendix A. SQL business questions and results

#	Business question	SQL technique	Headline result
1	Overall financial health and delinquency exposure	INNER JOIN, aggregates	10,293 customers; \$45.53M spend; \$11.98M revolving; \$10.98M income; 6.06% delinquency
2	Which card tiers earn the most relative to CAC?	GROUP BY, NULLIF ratio	Platinum \$27.75 and Gold \$24.46 per \$1 CAC vs Blue \$10.33
3	Quarterly trajectory of spend	COUNT DISTINCT, AVG	Spend \$11.1M to \$11.7M per quarter; Q4 \$11.70M with 2,689 accounts (14 weeks)
4	Which expense categories drive spend and ticket size?	Scalar subquery for share	Bills 24.5% of spend; Travel ticket \$100.51
5	Activation by acquisition contact channel	Conditional aggregation	Unknown 61.0%, Cellular 56.9%, Telephone 53.9%
6	Job and education segments with highest delinquency	HAVING (min 50), ORDER, LIMIT	Self-employed Doctorate 11.11%; Government Uneducated 9.13%
7	Utilisation across income tiers	CTE with CASE bands	70k to 100k tier: 36.5% utilisation; 468 accounts at 80%+
8	Payment channels: spend, ticket size and risk	CTE, SUM() OVER()	Swipe 62.6% of spend; Chip 7.24% delinquency
9	Does satisfaction relate to revenue?	DENSE_RANK() window	Score 1 ranks first (\$1,098) but spread is about 5%; no meaningful link
10	Pareto 80/20 spend analysis	Running total window function	5,856 cardholders (56.9%) generate 80% of spend